

Letter Pair Move Game

Time limit: 1.00 s

Memory limit: 512 MB

There are $2n$ boxes in a line. Two adjacent boxes are empty, and all other boxes have a letter "A" or "B". Both letters appear in exactly $n - 1$ boxes.

Your task is to move the letters so that all letters "A" appear before any letter "B". On each turn you can choose any two adjacent boxes that have a letter and move the letters to the two adjacent empty boxes, preserving their order.

It can be proven that either there is a solution that consists of at most $10n$ turns or there are no solutions.

Input

The first line has an integer n : there are $2n$ boxes.

The second line has a string of $2n$ characters which describes the starting position. Each character is "A", "B" or "." (empty box).

Output

First print an integer k : the number of turns. After this, print k lines that describe the moves. You can print any solution, as long as $k \leq 1000$.

If there are no solutions, print only "-1".

Constraints

- $1 \leq n \leq 100$

Example 1

Input	Output
3 AB. . BA	2 ABBA. . A. . ABB

Example 2

Input	Output
3 ABAB. .	- 1